



# Miguel Fernandes

Senior Data Scientist | AI | PhD in Neuroscience

✉ [amgfernandes@protonmail.com](mailto:amgfernandes@protonmail.com)  
☎ 0049-1523-7014457  
🌐 <https://www.neuroeduai.com/>  
🌐 [amgfernandes](#)

## Summary

Senior Data Scientist and PhD Neuroscientist with 10+ years of experience in data science. Expert in translating complex datasets into actionable business intelligence.

## Skills

🧠 AI      🤖 ML  
</> Git      🗄 SQL  
📄 Python

## Certifications

|                              |                                                |
|------------------------------|------------------------------------------------|
| Machine Learning with Python | Databases and SQL for Data Science with Python |
| April 2021                   | Python                                         |
| Coursera                     | May 2021                                       |
|                              | Coursera                                       |

## Languages

|                |          |
|----------------|----------|
| Portuguese     | German   |
| Native speaker | Advanced |
| English        | French   |
| Advanced       | Basic    |
| Spanish        |          |
| Basic          |          |

## Publications

All publications can be seen under <https://www.neuroeduai.com/publications>

## Experience

### BetterDoc

Cologne

Senior Data Scientist

October 16, 2022 - Present

- Built a GenAI assistant using advanced AI frameworks that allows non-technical teams to query complex corporate KPIs and business logic in plain language, significantly increasing data transparency and decision-making speed.
- Automated backend data mapping to connect raw database metrics directly to user-facing dashboards, ensuring the AI assistant delivers accurate, trustworthy answers.
- Analytics & ML: Python, SQL, Scikit-learn, XGBoost, Deep Learning, Predictive Modeling.
- Data Infrastructure & BI: AWS (Bedrock/Redshift), Postgres, dbt, DuckDB, Airbyte, Prefect, Apache Superset, Streamlit

### DZNE (Deutsches Zentrum für Neurodegenerative Erkrankungen)

Bonn

Head of the Image and Data Analysis Facility (IDAF)/Data Scientist

March 01, 2021 - September 30, 2022

- Led and mentored a 4-person team of Data Scientists and working students, streamlining research workflows and driving collaborative AI initiatives.
- Served as an internal data consultant, training multidisciplinary teams in advanced statistical analysis and data visualization techniques to improve organization-wide data literacy.

### Max Planck Institute of Neurobiology

Munich

Researcher/Data Scientist

September 01, 2013 - February 28, 2021

- Translated complex, high-dimensional datasets into clear, actionable insights to guide critical research decisions.
- Designed and deployed advanced machine learning algorithms (both supervised and unsupervised) to uncover hidden patterns and automate data analysis pipelines.

## Education

### University of Freiburg

PhD • 1.0

Neuroscience, Biology

Freiburg • April 01, 2009 - August 31, 2013

- Data analysis of imaging and time-series data and development of molecular and genetic tools for neural circuit dissection

## Awards

### Hans-Grisebach Award 2014

University of Freiburg

Best dissertation in the field of biochemistry and molecular biology in the year 2014

## Interests

Reading, Football, Biking, Cooking, Martial Arts